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APPENDICES

A.1 Source Code Manifest & GitHub Repository

ClearDrive AI is fully open-sourced under the permissive MIT license. Official GitHub Repository: **https://github.com/ShivaneshV/ClearDrive-AI** (Verified Production Commit: `767da5ad8519b429074cf9db9790676a7731322e`). The project repository is structured with complete automated reproduction scripts, regression tests, and pre-trained ONNX/PyTorch models.

File / Artifact	Size (Bytes)	Primary Technical Responsibility
`app.py`	54,603	Edge HTTP/HTTPS Web Server, MJPEG Video Streamer, GPS Speedometer, Watchdog Daemon.
`engine.py`	59,665	OmniVision Perception Core, Gyro Pitch Dampener, Ghost-Vision HDR, Pothole Scanner.
`dehaze.py`	15,273	Fast Guided Filter DCP Restoration, Temporal EMA Airlight, Foliage Floor Matrix.
`vehicle_tracker.py`	10,553	Pinhole Camera Ground Geometry, Distance Z (m), Closing Velocity, Dynamic TTC.
`dms.py`	12,811	MediaPipe 468-Point Facial Mesh, Micro-Eyelid EAR Microsleep & Inattention Sentry.
`templates/index.html`	48,200	Cyber Cockpit HUD, Dual-Panel Navigation Split, 0ms Local Trie Autocomplete Search.

A.2 12-Week Agile PBL Progress Log & Mentor Sign-Off Audit

Review Milestone	Date Held	Milestone Verified	Score / 20	Sign-Off Status
Review 0 (Ideation)	05 Aug 2026	Problem scoping, ODD formulation, requirements.	19 / 20	**APPROVED [Signed]**
Review 1 (Sprint 1)	19 Aug 2026	Sensor failure audit, 35-system literature survey.	18 / 20	**APPROVED [Signed]**
Review 2 (Sprint 2)	02 Sep 2026	Iteration 1 baseline, 1.8 FPS bottleneck profiling.	17 / 20	**APPROVED [Signed]**
Review 3 (Sprint 3)	16 Sep 2026	Fast Guided Filter & Tree Bug eradication.	19 / 20	**APPROVED [Signed]**
Review 4 (Sprint 4)	30 Sep 2026	OmniVision concurrency, Ghost-Vision & DMS.	20 / 20	**APPROVED [Signed]**
Review 5 (Final)	04 Oct 2026	Roadway validation, 55-page formal PBL report.	20 / 20	**RECOMMENDED [Signed]**

A.3 Self and Peer Assessment Quantitative Rubric

In conformance with the PBL continuous assessment framework, Table A.1 records the detailed evaluation rubric:

Evaluation Criterion	Max Marks	Shivanesh V (Lead)	Co-Researcher	Remarks & Evidence
1. Technical Mastery & Architecture	20	20 / 20	19 / 20	Decoupled multiprocessing worker pool; 38.6 FPS.
2. Physical Modeling & DSP	20	19 / 20	20 / 20	Fast Guided Filter & Tree Bug mathematical proof.
3. Real-World Road Testing	20	20 / 20	20 / 20	Ooty Ghat hairpin road trials & monsoon tests.
4. Documentation & Reporting	20	20 / 20	20 / 20	Exhaustive 55-page academic report.
5. Ethical & Open-Source Impact	20	20 / 20	20 / 20	MIT open-source release; Rs. 6,800 BOM safety.
Total Score / Final Grade	**100**	**99 / 100 (O)**	**99 / 100 (O)**	**Outstanding Performance**

Table A.1: Self and Peer Assessment Quantitative Rubric

A.4 Hardware Wiring & In-Vehicle Deployment Schematic

Electrical & Network Setup: 12V DC vehicle battery feeds an automotive buck converter stepping voltage to 5.1V DC 3.0A powering the Raspberry Pi 4B. Sony IMX323 dashcam connects via shielded USB 3.0 along the windshield headliner. The edge node broadcasts an encrypted private WPA3 Wi-Fi SSID (`ClearDrive-Core`), enabling in-vehicle Android screens to stream the 60 FPS HUD at `http://192.168.4.1:5000` with 28.4 ms latency.

A.5 REST API Telemetry Data Dictionary & Verification Sign-Off

Route	Method	Payload & Response Architecture
`/video_feed`	GET	Multipart/x-mixed-replace MJPEG video stream with AR HUD overlays at 30+ FPS.
`/telemetry`	GET	JSON: `{speed_kmh, safe_limit, ttc_seconds, collision_threat, ear_metric, driver_alert}`.
`/upload_frame`	POST	Accepts encrypted base64 JPEG buffers from wireless smartphone dashcam clients.
`/hud_toggle`	POST	Toggles active filter matrix modes: `{mode: 'dehaze'|'night'|'lanes', state: bool}`.

Verified & Certified by Faculty Guide:

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FACULTY GUIDE
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Endorsed & Forwarded by HOD:

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Dr. R. GOWRI, M.E., Ph.D.
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